

Wholesome Haven

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Introduction

Wholesome Haven is a gamified self-care mobile app that transforms everyday wellness tasks into interactive quests. By combining research-based self-care tasks with gaming elements, the app aims to increase users motivation, consistency, and emotional well-being.

Vision Statement

Wholesome Haven is a self-care gaming app designed for anyone who struggles to complete self-care tasks. Self-care tasks include everyday practices that support physical, mental, and emotional health... such as brushing your teeth, drinking water, journaling, stretching, or eating a balanced diet. These are small but essential actions that help people recharge and maintain balance. People have a hard time performing self-care tasks for various reasons: they have trouble remembering to do the tasks, are unmotivated, or don't know what the best self-care tasks are. Our app, *Wholesome Haven*, emphasizes self-care while using gaming elements, helping users stay motivated and healthy in an engaging way. This app will allow users to interact with friendly, non-player characters to receive "quests." Each quest corresponds to a real-world self-care task, connecting the concept of a "task" with the familiar gaming mechanic of a "quest." Based on completion of these quests, users will receive rewards, such as new plants in a garden, a new character moving in for the player to interact with, or items to decorate their house.

To ensure the self-care tasks are effective, they will be chosen based on research from books and scholarly journal articles. One article that can be used to direct research is Gamification for Health and Wellbeing: A Systematic Review of the Literature by Johnson et al. which examines gamification in the context of technology such as websites or apps (Johnson et al., 2016). The amount of new gameplay available in a day will be limited to prevent overuse of the app. At the end of the day, this application is aimed at promoting familiarity with and practice of activities that help users. If they leave with nothing but time taken out of their day, this app will have failed in achieving its goal. But if it helps them feel a little more cared for, proud of

themselves, and supported in building healthy habits, then *Wholesome Haven* will have done what it's meant to do... be a safe, wholesome space where people can learn to take care of themselves.

Background

There are many instances of technology being used in the mental-health field. The instances that are most relevant to our project are self-care apps and video games. Self-care apps and video games are examples that show how digital tools can help people build healthier habits, stay motivated, and feel supported in their everyday routines.

One popular self-care app is the Headspace meditation app by Headspace Inc. This app focuses on meditation and is designed to reduce stress and anxiety. In addition to meditation exercises, Headspace has podcasts, music and mindfulness exercises. What makes Headspace successful is how simple and approachable it feels - the calming visuals, sounds, and guided structure help motivate people to make good self care habits part of their every day routine instead of just another thing on their to-do list.

This next app has a wider array of uses. The Finch self-care pet by Finch Care Public Benefit Corporation features a bird companion that you can level up by completing self-care tasks. The tasks vary in type and difficulty and include things like drinking water, going for a walk, and taking out the trash. Users can customize which tasks they would like to focus on, making the experience specialized for each person's needs. Finch makes self-care feel personal and rewarding by letting users care for something that grows along with them.

A game used in a clinical context is EndeavourRx, a video game made by Akili Interactive Labs. EndeavourRx is an FDA approved video game for the treatment of ADHD in adolescents. According to EndeavourRx's about page, the game is for children aged 8-17 and must be prescribed by a doctor. For those not familiar with ADHD, in an article from December 2024, the National Institute of Mental Health described the three main symptoms of ADHD as inattention, hyperactivity and impulsivity. The goal of EndeavourRx is to move through a digital course and collect items while avoiding obstacles. The game is designed to challenge motor skills while also providing distracting stimuli. The distracting stimuli helps the user learn to multitask and ignore distractions. This shows how games can be designed with real intention and care - not just to entertain, but to help users grow new skills and strengthen their focus in meaningful ways.

Out of the aforementioned technologies our most direct competitor is the Finch self-care pet. It includes gamified self-care tasks much like our apps does. However, Finch is set up more

like a to-do list and less like a game than our app is. Wholesome Haven aims to fill that space in between. It keeps the heart of self-care apps while creating a more immersive and story-driven world that feels alive, comforting, and rewarding to return to. Wholesome Haven achieves a more game-like experience through the use of non-player characters, quests, a customizable house and in-game lore. Instead of receiving a list of tasks to complete, users of Wholesome Haven will be given quests that pertain to the game world.

As previously mentioned, the review titled *Gamification for Health and Wellbeing: A Systematic Review of the Literature* by Johnson et al. examines gamification in the context of technology such as websites or apps. It provides a basis for the combination of gamification of self-care tasks and digital technology to boost health. The review highlights how rewards, progress tracking, and story elements can make healthy habits feel more fun and sustainable which directly supports our design goals.

Another relevant article is *User experiences of college students using mental health applications to improve self-care: Implications for improving engagement* by Nagar et al. details what students liked and disliked when using various self-care apps (Nagar et al., 2023)c. This study shows us how important it is for self-care tools to feel gentle and personal. People are more likely to keep using an app that feels encouraging and calming rather than judgemental or overwhelming. The insights we have gained from studying competitor apps and research will help guide how we design both the look and feel of Wholesome Haven.

Normative and Ethical Considerations

As we designed and built Wholesome Haven, we were intentional about how each design decision would impact user experience. Self-care tools can become overwhelming or stressful if they're not carefully thought out, so in designing this, we were being careful about how everything works. Our goal was to make the app feel supportive, helpful, and kind - not discouraging, confusing, or heavy. That meant basing our decisions off of our design values on both the technical and creative ends.

One of our biggest priorities was keeping the app low-pressure while still encouraging engagement. We intentionally avoided systems that could create guilt, such as expiring quests or overwhelming task lists. Instead, users are able to move at their own pace, which helps the app feel encouraging. We also chose not to include comparison-based social features because even subtle competition can negatively impact mental health. By keeping progress private, the app maintains a more emotionally safe environment.

We also built flexibility into the quest system. Users are able to decline tasks because we recognize that people's energy levels, abilities, and circumstances change. This allows the app to better meet users where they are instead of forcing them into a fixed structure. Below are the key norms and how they were applied in the final implementation:

- **Cultural Appropriateness:** People come from many different backgrounds, and what feels like a positive self-care task to one person might feel uncomfortable or inappropriate to another. If these differences are ignored, users may feel left out or underrepresented. To address this, *Wholesome Haven* offers different types of self-care through distinct characters, including walking, nutrition, and cognitive restructuring. This allows users to choose what feels most realistic and supportive for them. Within each task, users are able to set their own goal, whether that be the amount of steps or servings of fruits and veggies. Users are also able to decline or adjust tasks as their needs change. This reflects the belief that self care is not one-size-fits-all. Each of us is made unique in God's image, and *Wholesome Haven* wants to care for and respect user differences.
- **Transparency:** We don't want users to misunderstand how *Wholesome Haven* should be used. It is designed to encourage healthy habits, not to replace therapy, medication, or other forms of care. Because of this, we were intentional in how the app communicates its purpose through both wording and design and we avoided language that suggests diagnosis or treatment. We also prioritize transparency in how data is handled. User progress is stored locally on the device, and no personal data is shared externally. If any data were to be shared beyond that, users would be informed.
- **Social:** Although social features can be helpful, they can also introduce pressure and comparison. We intentionally chose not to include features such as leaderboards, adding friends, or public progress tracking. This decision directly shapes the experience of the app. Users are focusing on their own growth instead of evaluating themselves against others. By removing comparison, the app supports a more internal sense of motivation instead of external validation.
- **Stewardship:** We designed *Wholesome Haven* to support users' well-being without competing for excessive time on the app or attention span. This influenced decisions such as limiting how many quests are available at once and avoiding systems that encourage excessive use. The app is meant to be used intentionally and then set aside to help form self-care habits in real life, not just within the app itself.

- **Aesthetics:** Aesthetics were a major part of how we shaped the user experience. All visual elements, including backgrounds, characters, and environments, are hand-drawn, creating a cohesive and whimsical style. This was a deliberate choice to make the app feel calm, personal, and approachable instead of clinical or overwhelming. The simplicity of the design also supports usability by helping users focus on the tasks themselves instead of figuring out a complex interface.
- **Justice:** We recognize that users have different abilities, resources, and starting points. Because of this, tasks are designed to be flexible which allows users to set their own goals and engage in ways that are realistic for them. For example, step goals and nutrition task amounts can be adjusted. There is not a fixed amount, and although the app may encourage you to up your step goal or serving size at some point, it is never forced. This ensures that the app allows users to define progress in a way that fits their lifestyle.
- **Caring:** Caring is at the center of how we designed *Wholesome Haven*. We didn't want to create something that adds pressure, comparison, or makes people feel like they're falling behind, especially knowing how apps can negatively impact mental health. Instead, we focused on keeping the experience low-pressure and encouraging, with small, manageable tasks that users can do at their own pace. The goal is to support users in everyday life, not to keep people on the app as long as possible.
- **Trust:** Trust is built by ensuring the app is both thoughtful and grounded in research. The self-care tasks included are informed by Evelyn's honors thesis, which guided how our tasks were selected and structured. This means the app is engaging and intentional in how it supports users. By combining research-backed practices with clear design choices, our app builds credibility and helps users feel confident in what they're using.

Product Success Criteria

- **Research Basis**
 - Minimum: passes user testing and receives generally positive feedback.
 - Stretch: develop a system for quantifying success and general parameters of good / bad goals to allow for a process of adding future goals.
- **User State Storage**
 - Minimum: stored locally on device allowing for privacy and continuity.
 - Stretch: cloud account for progress and changes to be tracked across devices.
- **Self-Care Interaction**

- Minimum: user should be able to figure out how to use it - fundamental high-priority task implemented.
- Stretch: more tasks implemented.

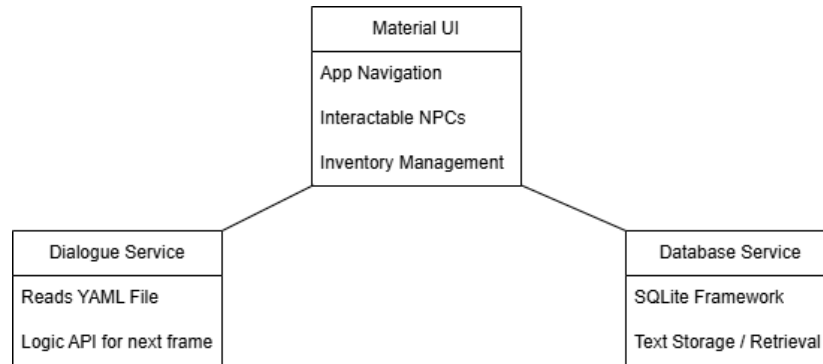
Approach and Implementation

Throughout the course of the project, we held weekly team meetings on Monday nights with our team and our advisor Keith VanderLinden. We also scheduled additional team meetings when needed, especially during the planning stages of our project since *Wholesome Haven* was built from scratch. Our team primarily communicated through discord where we discussed ideas, shared updates, and collaborated on different aspects of the development process. We also used Trello to organize tasks, assign responsibilities, and track progress throughout the semester.

During our meetings in the spring semester, we conducted three rounds of user testing with our advisor. The feedback we received from him helped us identify areas to improve and areas to implement changes across multiple areas of the app. Most of these changes were related to navigation, usability, and user experience. This process allowed us to refine the project as our development progressed. Because of the team's decision to hand-draw each asset, these UI changes were made while keeping a uniform, welcoming aesthetic.

From a technical standpoint, *Wholesome Haven* is built on Flutter, an open-source cross-platform framework that allows the source code to be compiled into both IOS and Android formats. Because there was no need for a social aspect, the team decided to keep all logic and data on the client side. This meant that a simple API for storing data in long term memory was necessary, taking session information and saving it to allow for retrieval during reboots. This functionality was built using SQLite, which is a library for embedding SQLite data storage into flutter applications.

The application also required large amounts of dialogue logic. Because of how the progression and text would be subject to change as research progressed regarding the proper structure of user quests, a YAML-to-dialogue service was created. This service takes the dialogue nodes defined within a YAML file and provides the logic for each dialogue with a Non-Player Character (NPC), allowing easy changes to quest structure. The UI takes the database and dialogue services to provide a user experience that follows research-based quests and allows for easy pick-up after the app has been closed.



Some of the major challenges that arose were from changing quest designs and flaws revealed from user testing. The database service was initially designed with external quests in mind. This meant that quests would accrue progress outside of the application and logged within the app. The number of steps taken and the mental reframing cycles would be encouraged by a quest, completed by a user, and marked by the app. This meant that a simple dictionary system would be suitable for tracking quantifiable progress metrics. All of the quest logic would be contained within the dialogue system that initiates and explains the quest. However, with the cyclical reframing quest for mental health, a chat interface was requested, which was outside of the capabilities of the dialogue service. As such, the database service was rewritten to allow for more flexibility.

Another challenge was UI usability. Each major iteration of the app faced a round of user testing where people were encouraged to interact with the NPCs and complete quests, but early rounds of user testing revealed that users had a hard time navigating the app and finding certain features. In order to improve usability, micro-rounds of UI updates were done with users in order to find suitable visual indicators of items being interactable.

Results

We developed a functional demo of Wholesome Haven that demonstrates the experience we set out to create.¹ Users are able to interact with multiple non-player characters (NPCs), receive self-care quests, and track their progress. The main features currently implemented include:

- **NPC Interaction:** Users can interact with three primary characters... Bob the Gnome, Shelly the Snail, and Clyde the Cloud, through a dialogue-based system. Each interaction includes options to accept, decline, or view progress on a quest.

¹ A video demo of our final product can be found here:
https://drive.google.com/file/d/1xPBZNw8c2cMazShB5IUQPRAUs_PdbarkK/view?resourcekey

- **Character-Based Quest System:** Each character is tied to a specific type of self-care task. Bob the Gnome focuses on fruit and vegetable intake, Shelly the Snail is focused on step-based goals, and Clyde the Cloud guides users through cognitive restructuring exercises.
- **Cognitive Restructuring Feature:** Unlike other quests, Clyde the Cloud utilizes a messaging-style interface. Users input their thoughts, and Clyde guides them through a series of questions to help reframe those thoughts. After the conversation finishes, users are able to scroll back up and see previous interactions.
- **Functional Step Tracking:** The step-based quest allows users to either manually input their steps or use a fully implemented step counter. This flexibility supports different user preferences while demonstrating integration between real-world activity and in-app progress.
- **Navigable Interface and Visual Design:** Our app includes a fully navigable interface full of entirely hand-drawn backgrounds, characters, and quest prizes. We chose to go with fully hand drawing each element to ensure they all match the whimsical design we were going for.

Together, these features show a strong and cohesive implementation of Wholesome Haven's storytelling, quest mechanics, and overall design. The app is fully functional, and the systems we built, especially the dialogue tree, make it easy to add and edit NPC interactions. This makes expanding the app manageable moving forward. Overall, this foundation shows how our creative vision and technical work came together and gives us a clear direction for how we could keep building and improving the app.

To evaluate the success of Wholesome Haven, we compared our current progress to the success criteria established at the beginning of the project.

- **Research Basis:** Our project met the minimum criteria. We conducted multiple rounds of user testing with our advisor and incorporated his feedback into the app's design. For example, based on his feedback about user navigation, we added visual indicators (such as exclamation points above NPCs) to signal interactive elements and help users know where to click. These improvements helped enhance usability.
- **User State Storage:** Our project successfully met the minimum requirement by implementing local storage of user progress within the app. This is shown in features such as the cognitive reframing quest, where users can scroll back and access their

previous interactions. By preserving user input over time, the app supports continuity, reflection, and a more meaningful engagement.

- **Self-Care Interaction:** Our project exceeded the minimum requirement. Users are able to navigate the app, interact with NPCs, and successfully complete three fully implemented self-care quests. These features demonstrate that the core gameplay loop is both functional and engaging.

There are several opportunities to continue improving and expanding on *Wholesome Haven*. One important area of future work is increasing cultural appropriateness and accessibility within our current self-care tasks. We recognize that people are created uniquely with different abilities, backgrounds, and needs. Because of this, a future improvement of the app could include alternative options for certain tasks. For example, for users who are unable to walk, an accessible alternative could be provided so that all users are able to participate in a way that works for them.

Another area of future work is expanding the variety of self-care tasks available. Although our current implementation has three fully developed quests, adding more quests would create a more personalized and engaging user experience. Additionally, there is room to refine the visual design of the app. Even though all the assets are hand-drawn and cohesive, future work could focus on polishing our visual elements and enhancing consistency across different screens and interactions.

From a technical perspective, future development could explore alternative data storage solutions. The app currently uses local storage but implementing cloud-based storage would allow users to save and access their progress across multiple devices. Overall, these improvements would allow *Wholesome Haven* to become more inclusive and flexible while continuing to support users in a meaningful and accessible way.

Conclusion

Wholesome Haven shows how gamification can make self-care feel more engaging and doable in everyday life. Throughout this year, we built a fully functional app where users can interact with three different NPC characters to complete three kinds of self care tasks, as well as reflect on their progress through both physical and cognitive activities.

One of the biggest things we accomplished this semester was creating a whimsical experience that connects everyday life to gaming in a way that feels natural and encouraging. Features such as the NPC dialogue system, the step counter quest, and cognitive reframing

with Clyde the Cloud show both the creative and technical aspects of our work. The structure we have created, especially the dialogue tree mentioned above, make it really easy to expand the app with new characters, quests, and features moving forward.

As a team, we each played a different role in bringing this to life. Musa and Landon mainly worked on the Flutter development and technical implementation of the app. Musa played a large role in the organization of software development, taking meeting information and creating concrete goals to be accomplished before the next checkup. On top of that, he developed many of the fundamental resources within the application such as the database API and UI templates, allowing for simple expansion of content as new assets and quest designs came in. His work initiated many of the screens and their navigation, on top of new screens such as the chat interface. In order to aid in this, he developed a codebase structure for members to put new quest logic and images in organized folders that were easy to maintain. He also created tools for simple debugging such as the database inspection screen for ease of development.

Landon worked more on the implementation and utilization of the app. Using many of the tools that Musa implemented, such as the database or screen widgets. He worked on the NPCs and their respective quests and UIs. This involved making a built-in pedometer to track the users' steps and rewards for quest completion, either being rewards inside the players' house or angry clouds becoming happy as more quests were completed. This meant he worked as a bridge between the hand-drawn elements and the back-end elements with work ranging from tracking total quest completions to placing the different UI elements across the app.

Evelyn and Madi focused on the design, research, and overall feel of the app. Evelyn created and refined the dialogue for the NPCs. She also did her honor's thesis on the gamification of self-care tasks to make sure the app included tasks and techniques that were proven to help improve the symptoms of anxiety and depression. Additionally, she directed the creative vision of the app to create a cohesive story that included the self-care tasks in an appealing way. She provided wireframe sketches to Madi in order to create the various screens for the app. She outlined desired uses of the app to help direct Musa and Landon in their Flutter development of the app.

Madi created the visual side of the app by hand drawing every NPC, background, and clickable detail seen on *Wholesome Haven*. She put a lot of intention into making everything feel cohesive, positive, and calming so the app feels welcoming and enjoyable to use. She also spent a lot of time working on the project presentation and final report to help clearly communicate the vision and progress of the app. Throughout the project, she worked closely

with the team, whether by brainstorming ideas with Evelyn and collaborating with Musa and Landon to help bring those ideas to life in the final product.

Overall, this project shows how although technology is not neutral, it can be used in ways that feel supportive, creative, and helpful. We have created a very strong foundation of *Wholesome Haven* that can be built on in the future. We would like to thank our advisor, Keith VanderLinden, for his continued support, guidance, and encouragement throughout this project. We are especially grateful that he stayed with us through the spring semester even after his retirement. We also want to thank our families and friends for their constant support and encouragement throughout this process. Additionally, we are grateful for each member of our team. This project would not have been possible without the collaboration, creativity, and effort that everyone brought to the table. Lastly, we thank God for the opportunity, ability, and inspiration to bring this project to life.

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